Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		iron(III) oxide		1		1		
		(ii)		award (1) for either of following this happens as a lower temperature than the melting point of aluminium oxide the melting point of cryolite is lower than the melting point of aluminium oxide accept lowers the melting point of aluminium oxide accept increases conductivity of the electrolyte neutral answer – lowers energy cost	1			1		
		(iii)		award (1) for either of following anode / graphite reacts with oxygen / air anode / graphite is burnt away / used up award (1) for either of following forms carbon dioxide $C + O_2 \rightarrow CO_2$	2			2		
		(iv)		$2Al_2O_3 \longrightarrow \boxed{4}\,Al + 3\,O_2$ balancing aluminium atoms (1) product (1) discrete marks		2		2		

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>award (1) for any of following
extraction (of aluminium) is very expensive
extraction is much more expensive than recycling
recycling is much cheaper than extraction

award (1) for reasoning
because much larger amounts of electricity/energy needed

award (1) for reference to conserving the ore if no other mark awarded</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td></td><td><table>
<tr><th>Left-hand side of the membrane</th><th>Right-hand side of the membrane</th><th></th></tr>
<tr><td>Na^+ Cl^- H^+ OH^-</td><td>Cl^- H^+ OH^-</td><td></td></tr>
<tr><td>H^+ OH^- Cl^-</td><td>Na^+ H^+ OH^-</td><td></td></tr>
<tr><td>Na^+ Cl^- H^+ OH^-</td><td>Na^+ H^+ OH^-</td><td>✓</td></tr>
<tr><td>Na^+ Cl^-</td><td>Na^+ OH^-</td><td></td></tr>
</table></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td><table>
<tr><td>titanium burns in chlorine in aqueous conditions</td><td></td></tr>
<tr><td>titanium doesn't react with chlorine under any conditions</td><td></td></tr>
<tr><td>aqueous conditions prevent chlorine from reacting with titanium</td><td>✓</td></tr>
<tr><td>aqueous conditions prevent chlorine from reacting with sodium chloride</td><td></td></tr>
</table></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
</table>

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(iii)		sodium hydroxide and magnesium hydroxide ✓ sodium hydroxide and sodium bromide ☐ sodium hydroxide and magnesium bromide ☐ magnesium chloride and sodium ☐			1	1		
				Question 5 total	**5**	**3**	**3**	**11**	**0**	**0**